Long-Horizon Task Planning for Autonomous Agents with Hybrid Architecture: a Systematic Mapping Study

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Abstract. The transition from rule-based systems to intelligent agents has introduced a "flexibility-compliance dilemma." While Traditional Robotic Process Automation (RPA) ensures compliance at the cost of flexibility, existing Large Language Model (LLM) based agents are adaptable but often struggle with long-term, complex tasks. This systematic mapping review examines 61 papers published from 2020 to 2025 to analyse publication trends, identify architectural mechanisms like neuro-symbolic integration, and reveal structural gaps in current autonomous planning frameworks.

Keywords: Autonomous Agents · Hybrid Architectures · Long-Horizon Planning · Large Language Models · Neuro-symbolic AI

1 Introduction

Thanks to predefined rules and meticulously designed workflows, traditional Robotic Process Automation (RPA), delivers significant value through cost reduction, improved efficiency, and error mitigation. For this reason, it has dominated the field of task automation for decades.

However, RPA systems tend to be rigid and fragile, making it difficult for them to adjust to changing environments or unforeseen user interface alterations [53].

The recent advances in Large Language Models (LLMs) have catalysed a paradigm shift toward cognitive automation, introducing a new class of "language agents". Contrary to traditional RPA, LLM–based agent are entrusted with the reasoning, planning, and decision–making required to achieve a goal. The inherent flexibility they have, due to their generalised knowledge capabilities, empowers them to tackle complex tasks with minimal tuning. This has historically been beyond the reach of traditional automated systems [12, 53, 68].

1.1 Problem Statement

The shift from strict, rule-based systems to adaptable agents has created a core trade-off: the "flexibility–compliance dilemma. Traditional RPA guarantees high

compliance at the cost of flexibility. At the same time, current LLM–based agents offer remarkable flexibility but suffer from significant deficits in reliability, especially as task complexity and horizon length increase [53]. Empirical evidence from benchmarks, such as *TravelPlanner*, shows that even state–of–the–art models like GPT–4–Turbo achieve success rates as low as 4.4%. Such failures highlight the main problem of LLMs, maintaining a coherent strategy over an extended sequence of actions [12].

These compliance issues arise from cognitive limitations in existing LLM architectures, as models struggle to reason over long contexts. Furthermore, they are notably weak in self–correction, often failing to recognize their own mistakes, and tend to produce confident yet incorrect outputs, a phenomenon known as hallucination. The research frontier aims to resolve this dilemma by developing cognitive architectures – multi–component frameworks that support and guide an LLM's reasoning. These systems explicitly differentiate between high–level planning and low–level execution, include feedback loops based on environmental reality for robust self-correction, and allow for dynamic replanning to handle unforeseen changes [12, 53, 68].

1.2 Research Objectives and Questions

The research goal of this study is to map and categorize the architectural strategies currently employed in the literature. Following the guidelines for systematic mapping [51], this study pursues the following specific objectives:

1. **Objective 1: Mapping the Research Landscape and Trends.** To systematically analyse publication trends, research types, and venues to understand the evolution of hybrid long–horizon planning.
2. **Objective 2: Identification of Challenges and Paradigms.** To identify the specific limitations current frameworks aim to address and the automation paradigms where these solutions are deployed.
3. **Objective 3: Synthesis of Mechanisms and Research Gaps.** To categorise the architectural mechanisms (e.g., Multi–Agent Systems (MAS), cognitive architectures) employed in the literature and identify existing gaps.

To achieve these objectives, this study addresses the following Research Questions (RQs):

RQ1: What is the trend in publication over time and the distribution of research types and venues in the field of hybrid long–horizon planning?

RQ2: What specific challenges and limitations in autonomous agents do current hybrid architectures aim to address, and what paradigms are they intended to automate?

RQ3: What architectural mechanisms are employed by MAS and agentic AI systems to address these challenges, and what are the identified gaps for long–horizon task automation?

2 Research Methodology

This study follows the guidelines established by Petersen et al. [51] to provide a reproducible overview of the researched landscape. The protocol aims to ensure descriptive and theoretical validity.

2.1 Search Strategy

This search strategy aims to identify relevant literature while managing noise. In order to achieve this, we utilised the PICO framework to define the study scope and derive the initial set of keywords for database–specific search strings.

PICO Framework. Comparison and Outcomes are not applicable to systematic mapping studies, as the goal is to map existing research rather than reviewing it [51]. The scope is defined as follows:

Population (P): Multi–agent systems, RPA systems, Software agents, Cognitive agents, Autonomous agents, LLM agents, Enterprise automation, Business process automation, Workflow automation, Long–horizon tasks, and Task automation.
Intervention (I): Planning challenges, LLM–based planners, CoT (Chain of Thought), Plan–Verify, Prompt engineering, In–context learning, Classical planning, HTN, PDDL, POP, Task decomposition, Goal reasoning, Cognitive/Hybrid architectures (BDI, Deliberative, Reactive), Persistent memory, RL, Self–correction, and Verification.
Comparison (C): Not applicable.
Outcomes (O): Not applicable.

Search String Development. The search strings were developed in a iterative process:

1. **Keyword Extraction:** Terms were extracted directly from the Research Questions and PICO sets.
2. **Synonym Identification:** These terms were expanded with synonyms and grouped into thematic sets.
3. **Refinement:** Combinations of these sets were iteratively refined through test searches to ensure the results were representative while remaining manageable.

Data Sources and Search Strings. The search was conducted in IEEE Xplore, Scopus, and the ACM Digital Library, because they index the most relevant and high–impact venues in the field [51]. The specific strings used for each database are summarised in Table 1.

Table 1. Search strings and database formats

Database	Search String
IEEE Xplore	`("Abstract": "autonomous agent" OR "software agent" OR "multi-agent" OR "bot" OR "agentic" OR "robotic") AND ("Abstract": "plan*") AND ("Abstract": "architecture" OR "hybrid" OR "cognitive" OR "neuro-symbolic" OR "neurosymbolic" OR "BDI" OR "belief desire intention")`
Scopus	`TITLE-ABS-KEY(("autonomous agent" OR "software agent" OR "multi-agent" OR "agentic" OR "bot" OR "robot*" OR "RPA" OR "robotic process automation") AND ("symbolic* plan*" OR "classic* plan*" OR "symbolic* reason*" OR "classic* reason*") AND ("architecture" OR "hybrid" OR "cognitive" OR "neuro-symbolic" OR "neurosymbolic" OR "BDI" OR "belief desire intention"))`
ACM	`(Abstract:("autonomous agent") OR Abstract:("software agent") OR Abstract:("multi-agent") OR Abstract:(agentic) OR Abstract:(bot) OR Abstract:(robot*)) AND (Abstract:(plan*) OR Abstract:(reason*)) AND (Abstract:(architecture) OR Abstract:(hybrid) OR Abstract:(cognitive) OR Abstract:("neuro-symbolic") OR Abstract:(neurosymbolic) OR Abstract:(BDI) OR Abstract:("belief desire intention"))`

2.2 Study Selection

The study selection process followed a multi–stage filtering pipeline to narrow down the initial search results to the primary studies analysed in this work.

Selection Criteria. The selection criteria were developed based on the guidelines of Petersen et al. [51]. The 2020–2025 timeframe was defined after analysing the initial data, which indicated that the volume of research in hybrid long–horizon planning did not emerge until 2020. Table 2 summarises these criteria.

Table 2. Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
Journals, conference papers, or preprints (this last one only if sampled from a peer–reviewed paper via snowballing)	Editorials, summaries, or keynote abstracts
Published between 2020 and 2025	Non–peer–reviewed or gray literature (unless sampled from a peer–reviewed paper via snowballing)
Focus on hybrid long–horizon planning related to RQs	Not focused on autonomous agent architectures
Language: English	Language other than English
Full–text availability	Duplicate studies of the same research

Selection Process. The search was performed on IEEE Xplore, ACM Digital Library, and Scopus, as they are known for covering most of the software engineering and computing literature. When available, filters were used to narrow results by publication type, language, year, and subject area to enhance relevance. Metadata such as titles, abstracts, and DOIs were exported for automated processing.

Preprocessing and Deduplication. The initial search produced 1,589 documents from IEEE, 212 from Scopus, and 1,045 from ACM. After applying the filters, the numbers decreased to 696 documents from IEEE, 98 from Scopus, and 298 from ACM. A Python script was used to normalise fields and remove duplicates from the combined dataset based on DOI.

Screening and Snowballing. An LM suggested exclusions if the documents were not in English or were not relevant for the topic. This suggestions where based on the title and abstract of the candidates. Finally, the suggestions were manually reviewed by the researchers to ensure reliability. Uncertainties were resolved through collaborative meetings. This stage resulted in 69 papers accepted for full–text screening.

During the full–text reading, 18 papers were discarded. For the first classified papers, those with the highest quality assessment score and highest relevance to the research questions were used to perform snowball sampling. This backward–and–forward sampling identified 15 additional papers. After the full–text evaluation, 5 were discarded, leaving a final set of 61 studies for extraction and classification. All accepted and rejected studies, with explicit reasons for exclusion, are included in the supplementary material for reproducibility. mp

2.3 Data Extraction and Quality Assessment

To ensure consistent interpretation across reviewers, a structured data extraction form was developed iteratively.

Extraction Form Structure. Each included paper was reviewed, and metadata was captured to systematically address the research questions. Ambiguities were resolved during weekly meetings to refine definitions. The extraction form is organised into the following segments:

- **Metadata:** ID, DOI, and Title.
- **RQ1 (Trends):** Year, Venue Name, Venue Type, and Research Type.
- **RQ2 (Context):** Primary Challenges and Target Automation Paradigm.
- **RQ3 (Mechanisms):** Agent Configuration, MAS Coordination, Neural Planning Mechanism, Symbolic Planning Mechanism, Memory Type, Self–Correction, Compliance Reasoning, and Task Horizon.
- **Quality Assessment (QA):** Clear Goal, Clear Method, Validity Discussed, Empirical Evidence, Results Reported, and Limitations Acknowledged.
- **Administrative Fields:** Extraction Notes, Second Review Flag, Snowballed, Selection Decision, and Reason for Exclusion.

Quality Assessment Strategy. Methodological rigour was evaluated using six quality dimensions adapted from Petersen et al. [51]. Each study was assigned a binary score (0 for No, 1 for Yes) for each criterion, as detailed in Table 3.

Table 3. Quality assessment criteria and scoring guide

Criterion	Description	Scoring Guide
Clear Goal	Assesses whether the study explicitly defines its research aims, goals, or specific RQs.	1 = Explicitly defined; 0 = Vague or missing.
Clear Methodology	Explains process (identification, selection, extraction, analysis).	1 = Explicit/reproducible; 0 = Incomplete/unclear.
Validity Threats	Identifies/reflects on threats (descriptive, theoretical, etc.).	1 = Discussed with mitigation; 0 = No discussion.
Empirical Evidence	Reports data from analysis, case studies, or validation.	1 = Empirical results present; 0 = Purely conceptual.
Clear Results	Findings are structured and interpretable.	1 = Organized (tables/figs); 0 = Poorly structured.
Limitations	Identifies constraints or areas for improvement.	1 = Explicitly stated; 0 = No mention.

Scoring and Interpretation. The Total Quality Score is the sum of all six criteria, ranging from 0 to 6. Studies are classified using the following customised categories, inspired by the ones defined in Petersen et at. [51]:

- **High Rigour (6):** Reliable for synthesis and comparison.
- **Good Rigour (5):** Included with caution.
- **Limited Rigour (0–4):** Treated as background evidence.

2.4 Classification Schemes

The analysis classified the selected studies using both topic–independent and topic–specific schemes.

Topic–Independent Classification. Each paper was categorised along the following dimensions:

- Research type: evaluation, validation, solution proposal, philosophical, experience, or opinion.
- Venue type: journal, conference, preprint.

Topic–Specific Classification. A domain–specific taxonomy was developed following keywording and an open coding process derived from full paper reading. While the detailed mappings for these schemata are provided in the supplementary material, the high–level definitions are as follows:

RQ2 [Challenges Addressed].

- **Long–Horizon Failure:** Challenges related to tasks requiring extensive sequences of actions or reasoning [8]. Includes issues with subgoal decomposition and managing agents in continuous state spaces [1, 34].
- **Computational Performance:** Limitations regarding the efficiency and scalability of the agent [10]. This includes real-time latency requirements [28], resource constraints in Industry 4.0 settings, and high training/execution costs.
- **Explainability:** The difficulty an agent has in offering rationales for its actions that humans can understand. It tackles problems like "hallucinations" in LMs [52], ambiguity, and the necessity for transparency, often through neuro-symbolic methods [24].
- **Safety and Compliance:** Risks associated with task execution in safety-critical environments. Focuses on preventing compliance violations [49], ensuring privacy, and implementing ethical reasoning or conduct governance [20].
- **Brittleness:** This refers to a lack of robustness where systems tend to fail when faced with scenarios outside their training data or predefined rules [67]. It highlights the need to address failures that occur in edge cases or when rigid rules are broken.
- **Unstructured Environments:** These involve difficulties in functioning within open-world or complex settings that lack a fixed schema. Such environments require semantic understanding, world modelling, and grounding of image schemas [41].

RQ2 [Automation Paradigms].

- **Embodied AI / Robotics:** Solutions explicitly designed for agents interacting with a physical or simulated 3D environment [32, 48].

- **Environment–Agnostic:** Solutions that address the fundamental logic of hybrid planning and reasoning without being tied to a specific environmental context. These architectures aim to solve the general "flexibility–compliance" dilemma [37, 54] by creating a universal framework that could be applied across diverse digital or physical domains.
- **Knowledge-Intensive BPM (Business Process Management):** Solutions tailored for structured, professional digital environments [9]. These focus on automating complex administrative, legal, or organisational workflows where strict adherence to high–level business logic, formal knowledge representations, and regulatory compliance is paramount [55].
- **Specialised Cognitive Domains:** A grouping of studies focused on niche applications requiring structured output. This includes Task–Oriented Dialogue (conversational reasoning) [26], Narrative Planning (consistent storytelling) [19], Social Event Prediction (modeling human/social dynamics) [56], and Automatic Workflow Generation (translating natural language intent into executable procedural scripts) [63].

RQ3 [Neural Components].

- **Foundational Models:** Solutions that leverage large–scale, pre–trained models as general–purpose reasoning engines. In hybrid systems, these components typically act as high–level semantic translators, transforming unstructured natural language instructions or visual scenes into structured symbolic specifications, such as PDDL goals [17] or LTL constraints [43]).
- **Generative and Predictive Models:** Neural architectures built to generate new data structures, forecast future environmental conditions, or carry out hierarchical task decomposition. These models are frequently used to "hallucinate" potential solutions or subgoals, which are subsequently checked thoroughly by a symbolic system component [16, 24].
- **Reinforcement Learning:** These models improve agent behavior by learning through trial–and–error within a specified environment. These neural components focus on policy learning and skill acquisition, either operating unconstrained or guided by symbolic boundaries to ensure that the learned behaviour remains within safety or logic guardrails [15, 32].
- **Discriminative and Perception Models:** Neural components are responsible for feature extraction, object classification, and spatial understanding. Their main function is to process high–dimensional sensory data (like vision or audio) to generate grounded state information that can be used by symbolic reasoning layer [22, 58].

RQ3 [Symbolic Mechanisms].

- **Classical Planning:** Search–based techniques generate a deterministic sequence of actions to reach a goal state from a known initial state. This approach uses defined operators and symbolic search algorithms, such as A* or MCTS [34], to guarantee the logical consistency of long–term plans.

- **Formal Logic and Solvers:** Declarative systems used for mathematical verification, constraint satisfaction, or logical inference. These components utilise formal languages (e.g., LTL [47], ASP [18]) and solvers (e.g., SMT [55]) to prove the feasibility of plans or to derive new knowledge from existing symbolic facts.
- **Knowledge Representation:** Solutions that utilize semantic structures, such as Knowledge Graphs [47] or Ontologies [58], to organize information about the world. These provide the agent with a stable, symbolic "grounding" that constrains neural outputs and ensures that decision–making is informed by structured, factual relationships.
- **Cognitive Architectures:** Frameworks inspired by human mental models, focusing on agency, belief–desire–intention (BDI) loops [11, 21], and deliberative reasoning. These architectures handle the agent's high–level "mental state," coordinating reactive behaviours with long–term goal management.
- **Reactive and Execution Control:** Programmatic structures that handle real–time task execution and lower–level control loops. This includes the use of state machines [29], Behaviour Trees [52], or mathematical motion constraints to ensure that the agent's physical or digital execution remains stable [59].

RQ3 [Memory Systems].

- **Working Memory:** Solutions that use volatile, short-term storage to keep the immediate context of a task. This mechanism is typically used to store temporary state variables, current user instructions, or ephemeral rules needed for ongoing execution cycle [34, 64].
- **Episodic Memory:** Solutions that maintain a historical record of particular past events and agent experiences. This allows the agent to reference a "trace" of previous steps to inform decision–making and prevent repeating past errors over an extended period of time [44, 65].
- **Semantic Memory:** Solutions involving the storage of factual, structured knowledge about the world that remains invariant across different tasks. This often includes formalised knowledge bases [33], ontologies, or "TBox/ABox" structures that provide the agent with a stable conceptual framework [55].
- **Procedural Memory:** Solutions concentrate on retaining skills, "how–to" sequences, or internal procedural rules. This mechanism frequently employs Domain Specific Languages (DSLs) [63] or database–backed scripts to specify the standard operating procedures that the agent needs to follow.
- **World Model:** Solutions that maintain a comprehensive internal representation of the environment's state. This enables the agent to perform internal simulations, or "imagination" cycles [23, 41], to predict potential outcomes and plan actions without immediate interaction with the environment.

RQ3 [Self–Correction Approaches].

- **Intrinsic:** Correction loops on agent's own thoughts or plans for consistency. This involves metacognitive activities that help detect logical gaps either beforehand or during execution [45].

- **Extrinsic:** Reactive correction based on external signals [61]. The agent executes an action, observes a failure or an unexpected result in the environment, and utilises this feedback to adjust its subsequent planning or behavior [47].
- **Formal Verification:** The application of hard–coded logical checkers or symbolic validators (e.g., PDDL validators [39,61]) that provide an objective "proof" of plan invalidity. These systems mathematically identify violations of the state–space or action–logic.
- **Multi–Agent Audit:** A decentralised correction mechanism where a specialised agent (an "Auditor" [38], "Judge" [2, 55], or "Critic" [31]) is responsible for reviewing and catching the mistakes of the primary planning agent through a separate validation layer.
- **Human–in–the–Loop:** Correction mechanisms that rely on human intervention to provide the corrective signal. The agent pauses or presents its plan for human review, receiving direct adjustment instructions to ensure successful task completion [33].

RQ3 [Compliance Reasoning Paradigms].

- **Logic and Rule–Based:** Compliance enforcement through hard–coded logical statements, BDI constraints [21], or symbolic rule–engines. These establish a deterministic boundary that the agent's actions cannot break.
- **Agent–Based Auditing:** A structural method where compliance is maintained by a specialised auditor agent [9]. This agent reviews the primary agent's proposed actions against rules or policies before execution is authorised.
- **Formal Verification:** Using mathematical proofs or exhaustive state–space searches to confirm compliance [37, 59]. These methods offer a theoretical guarantee that the agent stays within a compliant state throughout the task horizon.
- **Trace and Behavioural Monitoring:** An observational approach where compliance is evaluated after or during operation by analysing the agent's "traces" (thought logs or action history) [47]. This includes monitoring LLM reasoning processes for hallucinations or policy violations.
- **Human–in–the–Loop:** Manual supervision involving a human overseer who finalises compliance validation, ensuring adherence to safety and ethical standards not yet formalised in the system [65].

2.5 Validation Plan

To ensure the rigour and credibility of the systematic mapping process, this study follows the validation principles outlined by Petersen et al. [51]. Multiple validity dimensions are addressed to minimise bias and strengthen confidence in the findings.

Descriptive Validity Strategy

Goal. Ensure all data is recorded accurately and objectively.
Strategy. Use a predefined data extraction form to capture all relevant information.

Theoretical Validity Strategy

Goal. Ensure that the study captures what it intends to capture, minimising bias in study selection and data interpretation.
Strategies.

- Apply inclusion/exclusion criteria consistently using decision rules.
- Use backward–and–forward snowballing to mitigate the risk of missing relevant studies.
- Keep a log of excluded studies and reasons for exclusion.

Interpretive Validity Strategy

Goal. Ensure that conclusions drawn are reasonable and grounded in the data.
Strategy. To ensure that conclusions are objectively grounded, all findings are derived strictly from the data extracted during the classification process. Every statement and conclusion is supported by the necessary citations from the primary studies to minimise researcher bias and maintain interpretive transparency.

Repeatability Strategy

Goal. Allow other researchers to replicate the study.
Strategy.

- Fully document the search strategy, inclusion/exclusion criteria, data extraction template, and classification schemes.
- Archive datasets, search strings, and analysis scripts in an open repository.
- Clearly describe any deviations from the planned protocol.

2.6 Validity Evaluation

The high overall rigour (5.30 average) provides a stable foundation for identifying structural gaps. However, significant variance exists in the correlations between quality, publication venues, and study types, revealing insights into the field's current maturity.

Reporting Trends by Research Type. The assessment reveals that quality "penalties" are often a byproduct of the specific research paradigm. Philosophical papers and literature reviews often omit empirical evidence because, by definition, they focus on abstract theorising or synthesis.

A more critical finding is that nearly all instances of missing results and empirical evidence occur within *Solution Proposals*. Several papers [2, 16, 21] propose new hybrid architectures but fail to provide the structured data or procedural transparency required for autonomous verification. This identifies a risk where new architectures are being proposed faster than they can be rigorously documented.

Venue-Specific Rigor. The higher reporting standards in top-tier robotics forums, compared with the variability in venues like AAMAS, suggest that emerging **Solution Proposals** are being introduced faster than they can be rigorously documented in some communities. This identified risk is particularly prevalent in studies that fail to provide the procedural transparency required for autonomous verification.

The Validity Reporting Gap. A critical finding is the discrepancy between system limitation reporting (92%) and formal validity reflection (79%). While authors are largely transparent about the functional boundaries of their systems, they rarely reflect on the formal validity of their studies.

Validity and Generalizability. A consistent interpretive challenge is the assumption of broad applicability. Many top–performing studies meet rigorous standards in 3D simulation benchmarks but do not address whether their findings will translate to real-world physical environments. This underscores a possible disconnect between the logic in simulations and the complex, non-linear nature of actual physical deployment.

Implications for the Mapping Conclusions. The high score for methodological description (93%) ensures that findings like the *Logical–Safety Gap* and the *Contextual Fragility Gap* are documented clearly enough for the community to transition toward a more rigorous paradigm of "autonomous verification." These gaps represent genuine architectural frontiers rather than poor reporting.

Theoretical Validity Threats Following Petersen et al. [51], we prioritized peer–reviewed databases (IEEE Xplore, ACL, Scopus). However, in this field, a significant portion of research is released as preprints. By excluding non–peer–reviewed, this study may be missing emerging architectural patterns from late 2024 and early 2025 that have not yet completed the peer–review cycle.

Furthermore, our search string focused on "hybrid" and "planning" constructs. The notable absence of GUI–based hybrid agents in our results, despite their industry presence, suggests that software–centric agents may utilize different taxonomies not captured by our specific search string. This lets us think we may have created a terminological barrier.

Interpretive Validity Threats To manage the high volume of data, an LLM (Gemini 3 Flash) was utilised to assist with the initial data extraction. This introduces a threat: a neural model is used to classify research on neural model limitations. To mitigate this, a manual verification of all of the extraction fields was performed, cross–referencing the model's output against the primary study text to ensure accuracy.

There is a risk that our quality assessment is optimistic, as the rubric did not explicitly penalise the lack of real–world industrial validation, which is typical for the currently dominant **Solution Proposal** research type.

Descriptive Validity Threats Our findings show a significant surge in publications for 2025 ($n = 33$). Because this study was conducted while 2025 publications were still being indexed, the data represent a "snapshot" of the explosion phase. Additionally, to ensure descriptive consistency, papers extracted early in the process were re–reviewed against the final refined taxonomy.

Generalizability and External Validity Threats A total of 52.7% of the identified solutions are concentrated in **Embodied AI and Robotics**. Consequently, the identified *Temporal–Simulation Gap* may be a specific artefact of the robotics domain, where spatial grounding is paramount, and may not fully generalise to digital–only software agents.

Additionally, the study was restricted to English–language publications. This introduces selection bias, as significant contributions from major AI research hubs in non–English speaking regions, particularly East Asia, may be underrepresented in the mapping.

Conclusion on Validity Threats The finding that 90% of papers are **Solution Proposals** could be influenced by another selection bias; by searching for "architectures" and "mechanisms," the search naturally invites papers that propose new systems. While we explicitly screened for evaluation papers, the lack of dedicated **Evaluation Research** remains a credible finding about the field's current maturity, even after accounting for search bias.

3 Results

3.1 RQ1: Publication Trends and Venues

Trend of Research Facets The field is experiencing an exponential growth, rising from one paper in 2020 to thirty–three in 2025.

Solution Proposals account for approximately 90% of the total mapped literature. Notably, there is a complete absence of papers identified as pure **Evaluation Research** or **Validation Research**.

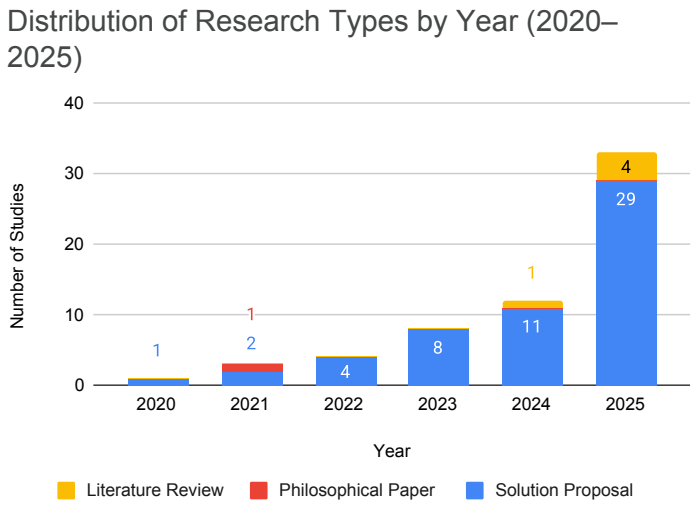


Fig. 1. Publication trends and research types in hybrid long-horizon planning (2020–2025).

Publication Venues and Distribution of Venue Categories AAMAS and ICRA emerge as the principal hubs for hybrid long-horizon architectures, together accounting for roughly 20% of the total papers identified.

Nevertheless, there are 35 different venues with only a single publication each. Hence, the application of long–horizon hybrid agents is distributed across diverse fields, including IoT [47], Human-Machine Interaction [38], and industrial software applications [23].

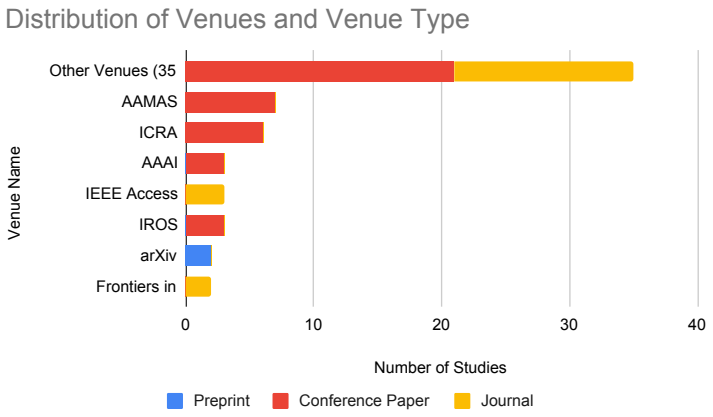


Fig. 2. Distribution of publication venues (2020–2025), highlighting the dominance of AI and robotics hubs.

3.2 RQ2: Challenges and Automation Paradigms

Automation Paradigms The research is concentrated in the areas of **Embodied AI/Robotics** and **Environment–Agnostic** frameworks, which represent nearly 90% of the mapped literature.

Within the scope of this systematic map, robotics appears to be the primary domain where the neural–symbolic split is functionally tested as it is the most frequent application context, accounting for 52.7% of the studies.

A substantial subset of the literature (36.4%) is classified as **Environment-Agnostic**. These studies propose architectures designed to solve the general logic of automation without being tailored to a specific physical or digital setting. Notably, **Knowledge-intensive BPM** represent only 3.6% of the papers.

The remaining studies in **Specialised Domains** (7.3%) cover niche applications such as narrative planning [66] and dialogue management [11]. They remain marginal to the core research effort, which is currently focused on broader challenges in robotic and generalised task execution. These results are inherently limited by the search string and inclusion criteria and reflect the state of the mapped literature rather than the full automation field.

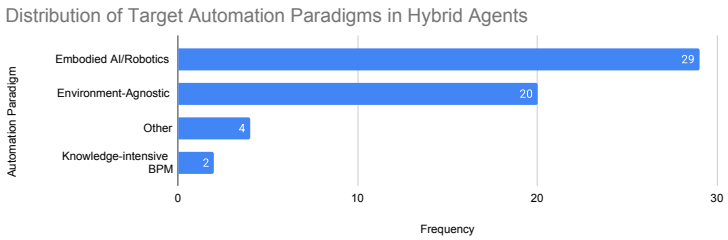


Fig. 3. Distribution of automation paradigms across identified solution proposals.

Challenges Addressed The research is being drived by five major drivers: **Explainability** [15,24], **Long-Horizon Failure** [8], **Unstructured Environments** [1], **Computational Performance** [10], and **Brittleness** [61,67].

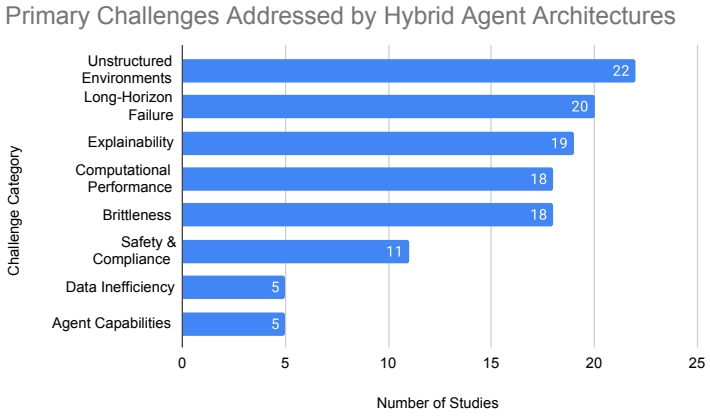


Fig. 4. Frequency of primary challenges addressed by hybrid architectures.

3.3 RQ3: Architectural Mechanisms

Neural Architectural Classes The distribution of neural components reveals a functional specialisation based on how stochastic models interact with the environment. **Foundational Models (FM)** emerge as the "generalist" component, demonstrating high affinity across **Explainability** (46.2%) and **Long-Horizon Failure** (38.5%) [34,36]. In the literature, FMs are primarily utilised as semantic engines that translate unstructured data into high-level abstractions for symbolic consumption [24].

In contrast, **Generative and Predictive Models** demonstrate a specialised role, with a 61.5% affinity for **Unstructured Environments** [18,65].

Similarly, **Discriminative and Perception Models** show a concentration in **Brittleness** (50.0%) and **Unstructured Environments** (41.7%) [22, 56]. These models exhibit a lower correlation with **Explainability** (15.4%) than other neural classes.

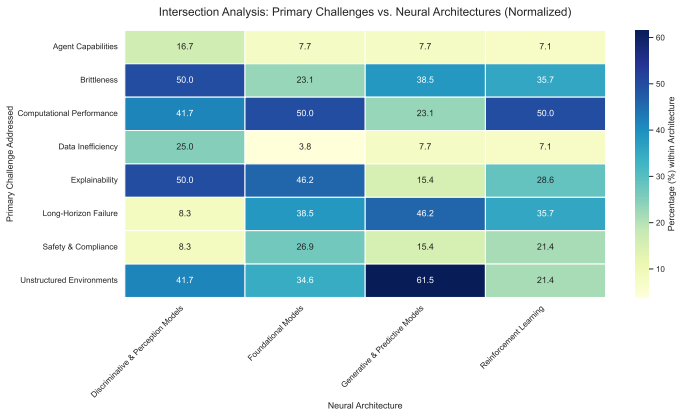


Fig. 5. Affinity heatmap between Neural Architectural Classes and identified research challenges, illustrating the distribution of model types across problem domains.

Symbolic Architectural Mechanisms Classical Planning serves as the primary tool for environmental management within the identified literature, showing its highest affinity for **Unstructured Environments** (59.1%) and **Long-Horizon Failure** (45.5%) [17, 26].

Cognitive Architectures (such as BDI [9, 11]) demonstrate a focus on **Explainability** (62.5%), with high–level mental models employed to address agent transparency [29, 38].

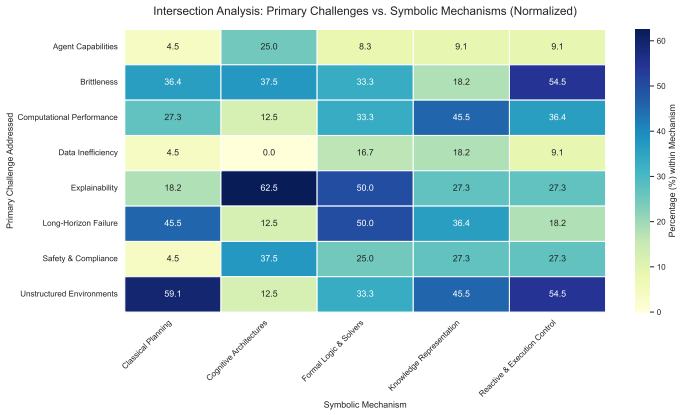


Fig. 6. Affinity heatmap between Symbolic Architectural Mechanisms and research challenges, showing the frequency of search-based and logic-based solutions.

Neuro-Symbolic Architectural Interfaces The data identifies significant pairings between:

- **Foundational Models** and **Classical Planning** [61].
- **Foundational Models** and **Formal Logic and Solvers** [43].
- **Foundational Models** and **Knowledge Representation**, often utilizing **Symbolic Knowledge Injection (SKI)** [8, 16].
- **Reinforcement Learning** and **Cognitive Architectures** [2].

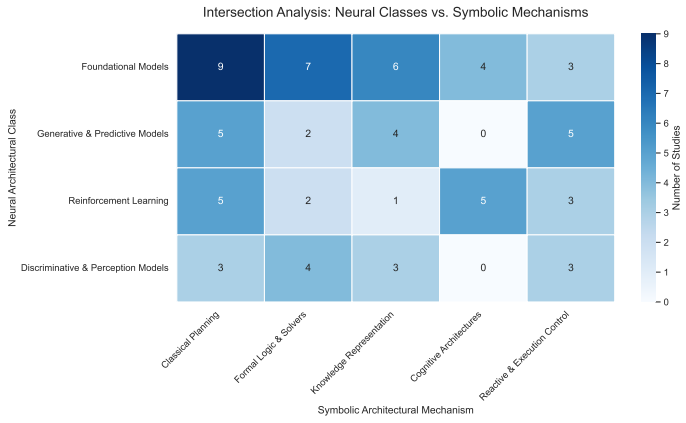


Fig. 7. Neural and Symbolic component pairings, identifying the most frequent hybrid architectural configurations in the literature.

Agent Configurations Single-Agent systems dominate the agent's configurations, accounting for 68% ($n = 38$) of the identified solutions. Normalised data reveals distinct functional specialisations across both configurations. **Multi-Agent Systems (MAS)** demonstrate a high affinity for **Explainability** (50.0%), **Single-Agent** configurations show higher correlation with **Brittleness** (36.8% vs. 22.2%) and **Unstructured Environments** (47.4% vs. 22.2%).

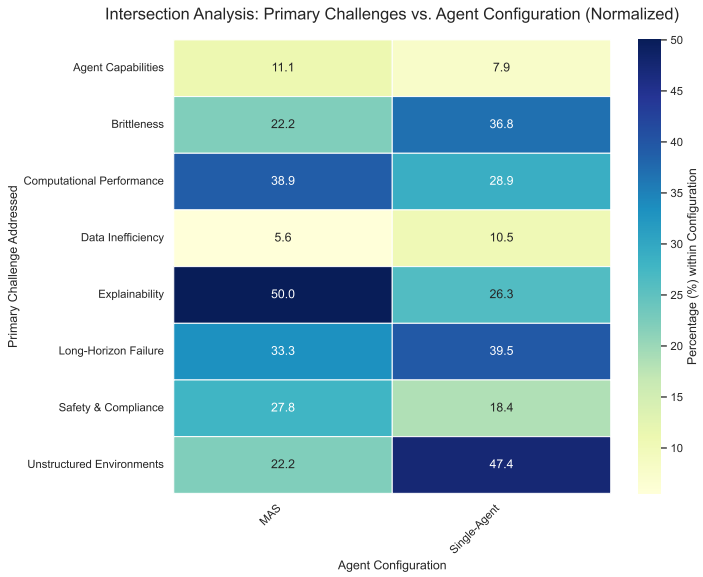


Fig. 8. Normalised affinity heatmap between Agent Configurations and Research Challenges.

Memory Types The analysis of memory systems identifies **Working Memory** as the main baseline across different architectural classes.

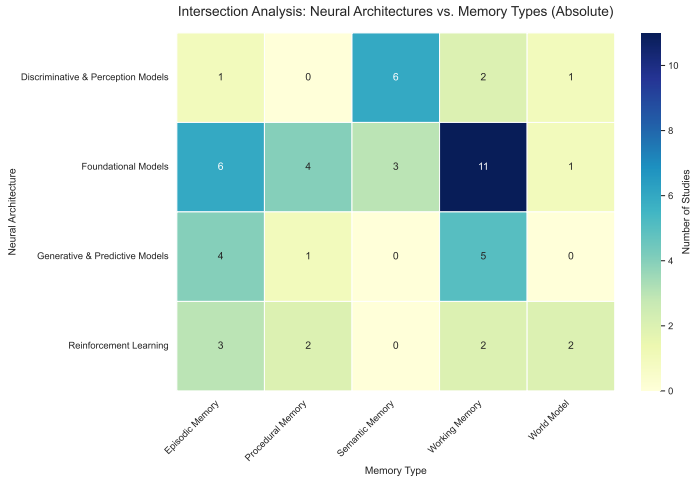


Fig. 9. Absolute frequency distribution of Memory Types across Neural Architectural Classes.

Foundational Models (FM) demonstrate a high utilisation of **Working Memory** (68.8% within the memory type), with roughly 40% of cases tethering these models to volatile context.

Semantic Memory shows its strongest affinity for **Explainability** (55.6%). A significant intersection is identified regarding **World Models**, which demonstrate a 100% affinity for both **Unstructured Environments** and **Brittleness**, while showing 0% affinity for **Long-Horizon Failure**.

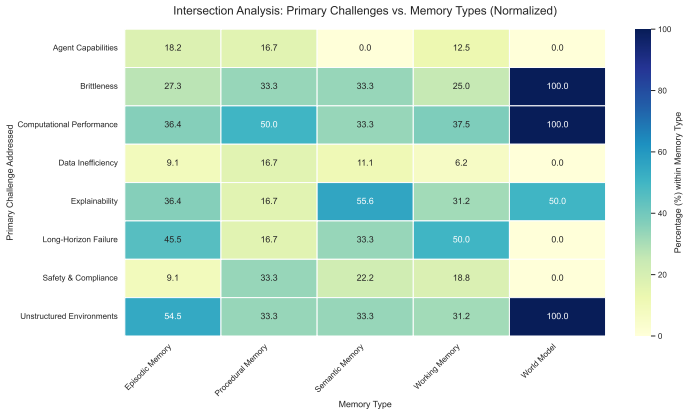


Fig. 10. Normalised affinity heatmap between Memory Types and Challenges, illustrating the alignment between structured memory and task automation constraints.

Self-Correction Mechanisms The mapping of self-correction strategies reveals different preferences based on the primary challenge addressed. **Intrinsic Reflection** is the most frequent mechanism used for **Brittleness** (52.9%) [45], while **Multi-Agent Auditing** is predominantly employed for **Explainability** (60.0%) [5, 30].

Human-in-the-Loop (HITL) remains the primary standard for addressing **Safety and Compliance**, it accounts for roughly 50.0% of the cases associated with this challenge [37].

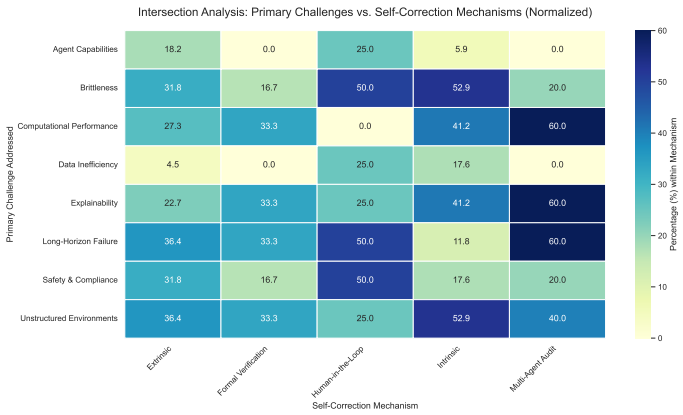


Fig. 11. Normalised affinity heatmap between Self-Correction Mechanisms and Challenges, illustrating the frequency of intrinsic reflection, multi-agent audits, and human-in-the-loop interventions.

Compliance Reasoning As illustrated in Fig. 12, the **Logic and Rule-Based** approach is the most common. It is commonly used for **Unstructured Environments** ($n = 20$) and **Long-Horizon Failure** ($n = 16$). Conversely, other paradigms like **Formal Verification** [60], **Agent-Based Auditing** [31], and **Trace and Behavioural Monitoring** [47] are much less frequently represented, with many challenge intersections showing zero or near-zero counts.

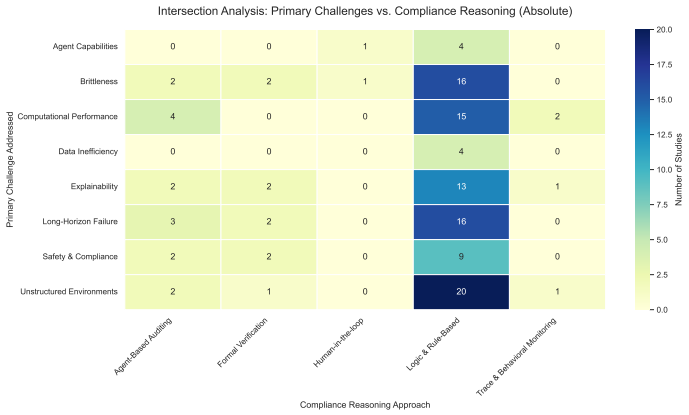


Fig. 12. Absolute frequency distribution of Compliance Reasoning Paradigms across Challenges.

3.4 Quality Assesment

To assess the reliability of the 61 identified studies, a 6–point scoring rubric was used, adapted from the dimensions outlined by Petersen et al. [51]. The field generally exhibits a high standard of reporting, with an average rigour score of 5.30.

Rigour Distribution and Venue Performance. The distribution of studies across rigour categories is as follows:

- **High Rigour (6/6):** 41 studies (67.21%).
- **Good Rigour (5/6):** 8 studies (13.11%).
- **Limited Rigour (<5/6):** 12 studies (19.67%).

While some high–tier venues, such as AAAI, IROS, consistently achieve a quality score of 6.00, others such as AAMAS show an average score of 4.38.

Analysis of Quality Criteria. The studies show excellent adherence to defining clear goals (100%) and describing the methodology used (93%). Threats to validity were identified in 79% of the papers, and 92% of the authors explicitly recognised system limitations, consistent with descriptive validity requirements [51]).

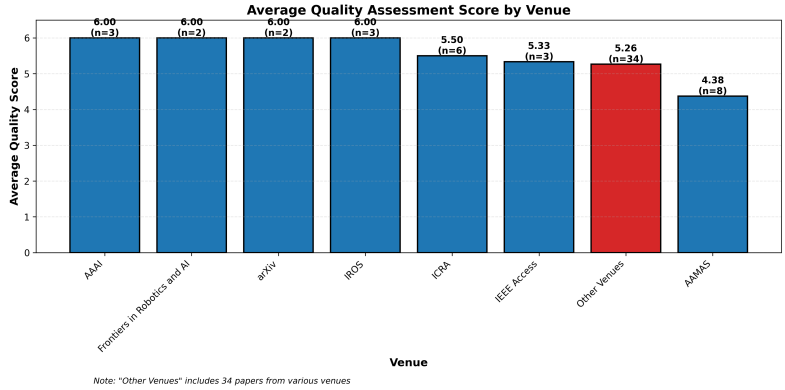


Fig. 13. Mean quality scores across primary publication venues and research types.

4 Discussion

4.1 Interpretation of RQ1: The Pace of Architectural Innovation

A Solution-Oriented Research Environment. The absence of **Evaluation Research** and **Validation Research** suggests that the field has not performed comprehensive evaluations on all the proposed solutions nor real–world validation.

Additionally, the lack of **Philosophical Papers** after 2021 indicates that researchers are moving away from abstract theorising towards the design of practical architectures for specific problems. In this context, the "theory" is likely integrated directly into the **Solution Proposals** themselves, signalling a pragmatic, implementation-oriented approach to tackling the long–term challenge.

A Fast–Paced Research Environment. The field is focused on engineering practicality, enabling researchers to quickly showcase tangible implementations and prioritising rapid sharing over the longer review times typical of journals. This can be infered by the 2:1 ratio of conference papers to journals. Hence, the idea that current theoretical work is mostly presented within architectural prototypes rather than as standalone philosophical concepts is reinforced by this arguement.

The Need for Consolidation and Synthesis. The emergence of **Literature Reviews** in 2024 and the rise in 2025 indicate that the volume of primary research has grown enough to require consolidation. This marks a transition from merely exploring to synthesising, in which the community seeks to develop shared taxonomies and establish standard benchmarks for hybrid agents.

4.2 Interpretation of RQ2: Domain Disconnects and Flaw Mitigation

The Dominance of Robotics within the Dataset. The high representation of **Embodied AI/Robotics** in this dataset likely stems from the specific needs of the physical environment. Robotic tasks usually demand a complex perception system to understand unstructured surroundings and a reliable planning component to perform tasks safely and logically [64]. In this systematic map, robotics is evidently the main area where the neural-symbolic division is tested against real–world constraints.

The Disconnect with Enterprise Settings. Considering the initial motivation of this research—shifting Enterprise RPA towards cognitive automation—the data suggests a disconnect between industry ambitions and current academic research. While balancing flexibility and compliance remains crucial for the future of RPA [53], the exploration of long-horizon hybrid agents for complex **Knowledge-intensive BPM** is notably less developed than for robotic or more general applications.

Fixing Fundamental Flaws. The high frequency of **Explainability** and **Long-Horizon Failure** challenges provides a clear justification for shifting towards hybrid architectures. This indicates that researchers mainly leverage symbolic components to offer formal logic and deterministic planning [66], which neural models often struggle with, while using neural components to manage high-dimensional perception required for **Unstructured Environments** [1].

Moreover, the lower counts for **Agent Capabilities** and **Data Inefficiency** imply that the field's current focus is not on adding new features to agents but on addressing fundamental issues of reliability and transparency. The presence of **Environment-Agnostic** solutions further reflects academic interest in modularity, considering the neural-symbolic interface as an independent architectural challenge.

Dual Planning Approach. A significant portion (roughly 41%) of robotics–centered papers focus on **Computational Performance**. This can be explained by a growing trend toward dual-planning architectures that mimic human cognitive processes: utilising fast but potentially unreliable neural components for immediate reactions and slow but reliable symbolic components for strategic planning [13].

4.3 Interpretation of RQ3: Mapping the Structural Gaps

Established Neuro-Symbolic Design Patterns

The Translation-Based Front-End. As shown in **Fig. 7**, the most common architectural interface in the dataset is the coupling of **Foundational Models**

(FM) with **Classical Planning**, representing a standardised **Translation-Based Front-End**. In this setup, the neural component acts as a semantic engine rather than just a perception layer [61], translating unstructured natural language or visual intent into a formal symbolic action language [17]. This design reflects an architectural strategy to reduce the unreliability and "hallucination" risks associated with large-scale neural models; researchers use the neural layer for its semantic flexibility, while the symbolic planner ensures logical consistency and supports long–term planning.

The Generative-Verification Loop. The combination of **Generative and Predictive Models** with **Formal Logic and Solvers**, also shown in **Fig. 7**, demonstrates a **Generative-Verification Loop**. In this pattern, the neural engine's flexibility is used to generate potential solutions, which are then checked against mathematical guardrails or temporal logic constraints [43]. The frequent appearance of this pairing in the literature indicates a strong emphasis on formal correctness and verifiable compliance in systems that incorporate generative neural engines.

Scrutability in Policy Optimisation. The clustering between **Reinforcement Learning** and **Cognitive Architectures** identifies a distinct pattern for achieving **Scrutability in Policy Optimisation**. This mechanism implies an integration in which symbolic frameworks are utilised to manage an agent's "mental state" and intentions as it learns policies through interaction [11]. This suggests that when neural models are utilised for policy optimisation, researchers frequently employ cognitive symbolic structures to provide a level of intentionality and transparency that neural weights alone cannot provide [2].

Mid-Level Orchestration. The presence of **Generative and Predictive Models** in both **Classical Planning** and **Reactive and Execution Control** highlights their function in **Mid-Level Orchestration**. These models serve as intermediaries, dividing complex tasks into smaller, manageable subgoals [24]. These subgoals are then handled by either high-level symbolic search algorithms for strategic planning or by real-time control loops for immediate execution, effectively linking abstract goals with physical or digital action [1].

Component Specialisation and Configuration Roles

Environmental Modelling Specialisation. The distribution of neural components reveals a specialised function in how stochastic models connect with physical or digital environments. **Generative and Predictive Models** are 61.5% affine for **Unstructured Environments** (**see Fig. 5**), suggesting a pattern where these models are used specifically to manage environmental ambiguity and state estimation [8, 65]. Similarly, the concentration of **Discriminative and Perception Models** in **Brittleness** (50.0%) and **Unstructured Environments** (41.7%), matching their role in approximating the current state via non-linear

sensor fusion [44]. Their correlation with **Explainability** (15.4%) probably reflects their connection with **Semantic Memory** structures rather than innate transparency in neural mechanisms [56].

Symbolic Roles in Temporal Consistency and Intent. In the symbolic layer, **Classical Planning** remains the preferred approach for maintaining logical consistency over extended timeframes, as shown by its high affinity for **Long-Horizon Failure** (45.5%) in **Fig. 6**. By using search-based algorithms, architectures reduce sequencing failures that neural models alone struggle with [17]. In contrast, the sharp specialisation of **Cognitive Architectures** in **Explainability** (62.5%) suggests that high-level mental modelling is mainly employed to provide intentionality [9, 29]. This implies that symbolic structures are seen as necessary to make complex agent decisions understandable to human observers.

Agent Configuration and Correction Paradigms. The dominance of **Single-Agent** systems (68%) likely reflects the simpler implementation of centralised architectures. However, the significantly higher affinity of single-agent configurations for **Brittleness** and **Unstructured Environments**, shown in **Fig. 8**, implies that the fragility of integration is viewed as an internal architectural failure [24]. This is echoed in the preference for **Intrinsic Reflection** (52.9%) to prevent failures internally (**see Fig. 11**) [45]. Conversely, **Multi-Agent Systems (MAS)** are prioritised for **Explainability** (50.0%) and **Multi-Agent Auditing** (60.0%), suggesting that researchers use role-based separation to address the "black-box" nature of neural models through external oversight and explicit communication traces [55].

The Human-in-the-Loop Boundary. Despite the push for automated self-correction, **Human-in-the-Loop (HITL)** remains the clear standard for addressing **Safety and Compliance**, as shown in **Fig. 11**. HITL serves as a reliable fallback when automated symbolic reasoning reaches its limits, particularly in high-stakes scenarios. This identifies a significant boundary in current autonomous capabilities, where human intervention remains the only reliable guarantor of verifiable compliance [37].

Identified Architectural Gaps

The Contextual Fragility Gap. **Foundational Models** largely operate without durable anchors, relying instead on the transient window of the prompt (**Working Memory**) in approximately 40% of cases. This utilisation pattern, identified in **Fig. 9**, highlights a potential gap in understudied ensembles that decouple long-term planning from the FM context window. Without persistent anchors, these architectures face a significant risk for extended task durability, as the agent's "memory" remains tethered to volatile context rather than stable, long-term symbolic structures.

The Logical-Safety Gap. Although safety is a key driver of hybrid architecture adoption [5], the distribution reveals a notable gap: symbolic layers are primarily used for logical task execution, with safety often viewed as a secondary outcome (**see Fig. 12**). This disconnect is further underscored by the overlap between symbolic techniques and compliance reasoning; the dependence on broad **Logic and Rule-Based** systems across various symbolic methods shows that the industry's need for fully reliable compliance is being addressed with general-purpose logic rather than specialised safety-critical reasoning.

Additionally, the 50% rate of explainability in **MAS** indicates that researchers prefer using role-based decomposition as an external audit method rather than an internal limitation. By introducing an "Auditor" agent to supervise an "Actor," compliance is maintained through decentralised monitoring and behavioral traces, offering oversight that the symbolic layer alone cannot yet guarantee [5, 30]. This reinforces the hypothesis that the fragility in neural-symbolic integration is that it results from internal failures of the individual agent, which should be addressed with external oversight.

The Temporal-Simulation Gap. Internal simulation is currently employed as a "reactive" tool, correcting perception in the immediate environment, rather than a "proactive" tool for simulating future sequences. While **World Models** show 100% affinity for **Unstructured Environments** and **Brittleness**, they demonstrate a 0% affinity for **Long-Horizon Failure** in the normalized data (**Fig. 10**). Without this temporal dimension, the self-correction of hybrid agents remains limited to immediate error recovery (via extrinsic feedback) or error prevention (via intrinsic reflection), rather than long-term strategic prevention through future-state simulation.

The Cognitive Modelling Gap. A notable structural gap exists in the absence of direct interaction between **Cognitive Architectures** and both **Generative/Predictive Models** and **Discriminative/Perception Models**. This gap indicates that high-level mental modeling is mainly mediated by planning or logic layers and seldom directly interfaces with raw perceptual processing or generative loops. These findings describe current neuro-symbolic engineering as primarily focusing on "Translation," "Grounding," and "Verification" pipelines, with limited integration of deep cognitive frameworks and neural sensing unexplored.

The Flexibility-Compliance Dilemma. The primary challenge of balancing flexibility and compliance is well recognized in the literature. However, the mapping indicates that peer-reviewed studies have yet to develop consistent patterns or specific reasoning methods to address this issue, aside from conventional symbolic constraints. Existing architectures tend to focus on the flexibility of generative neural engines, often viewing compliance as a secondary outcome of the execution process rather than a fundamental architectural feature priority.

5 Conclusion

This study maps the existing research on long-horizon task planning and evaluates how emerging hybrid architectures tackle the "flexibility-compliance dilemma" [37, 49]. Following established guidelines [51], our summary of evidence reveals a domain with high maturity in translation–based patterns and explainability. The literature shows that the domain remains hampered by structural fragility in terms of temporal durability and autonomous compliance. By examining the distribution across neural, symbolic, and architectural mechanisms the following gaps are identified: the Contextual Fragility Gap (the vulnerability of strategic planning to volatile or limited neural memory), the Logical-Safety Gap (the difficulty in ensuring strict rule adherence and operational compliance), and the Temporal-Simulation Gap (the absence of robust mechanisms to simulate and verify extended action sequences).

The findings demonstrate that while the research community has frequently proposed mechanisms to make agents more scrutable and adaptable, primarily through foundational model translation [21], the transition toward truly autonomous and reliable agents represents a critical and ongoing research frontier. The mapped architectures predominantly utilise symbolic components for immediate task sequencing. To maintain logical safety, the systems rely heavily on multi-agent auditing frameworks [5] and, to a lesser extent, human-in-the-loop fallbacks [69]. However, these methods still cannot provide a complete or robust solution for autonomous compliance.

These gaps yield a roadmap for the evolution of cognitive automation. Specifically, moving beyond reactive grounding and decoupling planning from volatile context windows represents an underexplored avenue. Furthermore, while the current literature frequently treats symbolic logic as an execution tool, evolving it into a robust framework for continuous autonomous verification requires further empirical investigation [28]. These gaps provide a strong foundation for future studies that aim to demonstrate that architectural flexibility and strict control can coexist.

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